



HACKSKILL
H2S

BHARATIYA ANTARIKSH HACKATHON 2026



Team Name : **WarpDrive**

Team Leader Name : **OMPRAKASH BEHERA**

Problem Statement : **PS-7 | AI-Enabled Detection of Exoplanets from Noisy Astronomical Light Curves**

Team Members

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● **Team Member**

Name: **Sujoy Bose**

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Problem Understanding & Opportunity

The Challenge

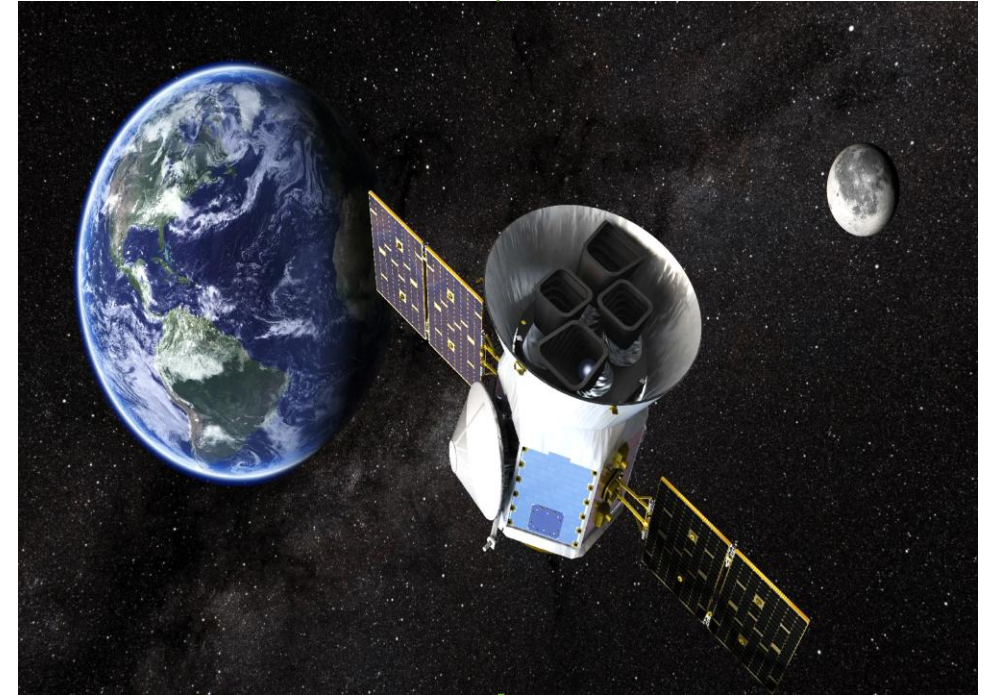
Exoplanet detection requires identifying **tiny brightness variations** in stars via transit photometry.

Key Complications

- Stellar blending in crowded fields
- Detector noise and systematics
- Confusion with eclipsing binaries & starspots

The Opportunity

TESS provides **20-30k light curves/sector**. We propose an **AI-driven end-to-end pipeline** to automate detection, classification, and characterization of transit signals.



NASA TESS Mission - Scanning the skies for exoplanets

[Problem Diagram Link](#)

[Distinction](#)

Key Features & Capabilities

Data Processing

- Automated detrending
- Noise & outlier removal
- Systematics correction

Parameter Estimation

- Transit depth calculation
- Duration & period refinement
- MCMC uncertainty quant

Detection & Period

- Periodic dip detection (BLS)
- Orbital period estimation
- Phase folding & alignment

Confidence Scoring

- SNR-based significance levels
- Model confidence intervals
- False positive probability

AI Classification

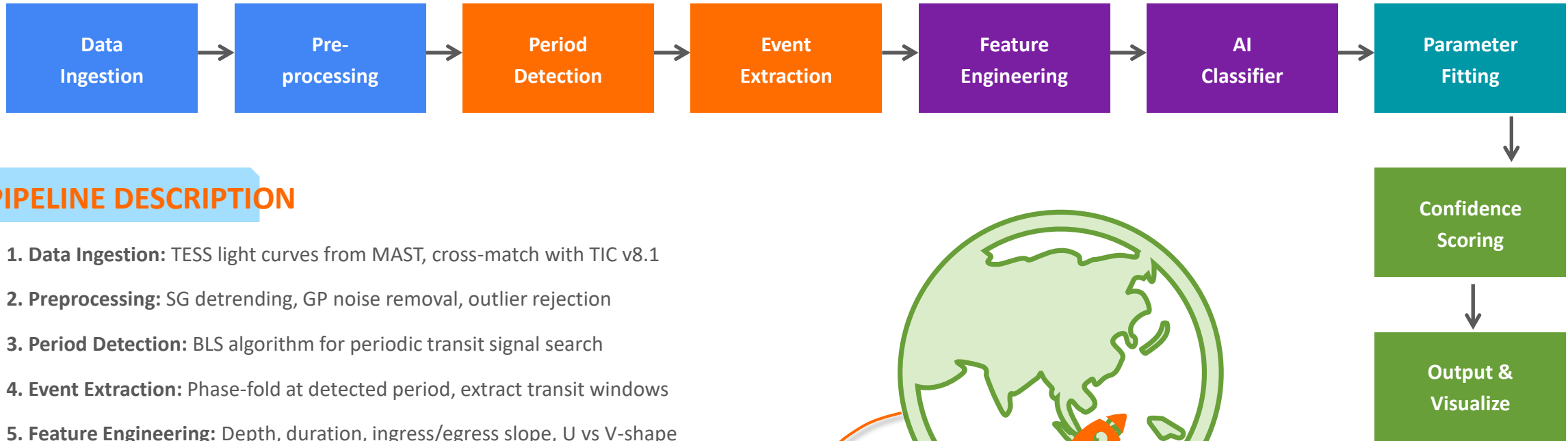
- Transit vs Eclipse vs Blend
- Shape feature extraction
- XGBoost / Neural Network

Visualization

- Interactive light curve plots
- Detected signal overlay
- Results dashboard

[Feature Architect Diagram Link](#)

AI Pipeline Process Flow



PIPELINE DESCRIPTION

- 1. Data Ingestion:** TESS light curves from MAST, cross-match with TIC v8.1
- 2. Preprocessing:** SG detrending, GP noise removal, outlier rejection
- 3. Period Detection:** BLS algorithm for periodic transit signal search
- 4. Event Extraction:** Phase-fold at detected period, extract transit windows
- 5. Feature Engineering:** Depth, duration, ingress/egress slope, U vs V-shape
- 6. AI Classifier:** XGBoost on confirmed planets, EB, blends, false pos
- 7. Parameter Fitting:** batman/LMFIT fitting, emcee MCMC uncertainties
- 8. Output:** SNR scoring, classification labels, parameter estimates



[Visual Pipeline with Data Flow Link](#)

TESS Data & Catalog Understanding

Data Source

TESS Full Frame Images (FFI) and 2-min target pixel files from [MAST Archive](#)

CTL v8.01 x TIC v8.1 Catalog

9.5GB CSV with 100+ stellar parameters per source:

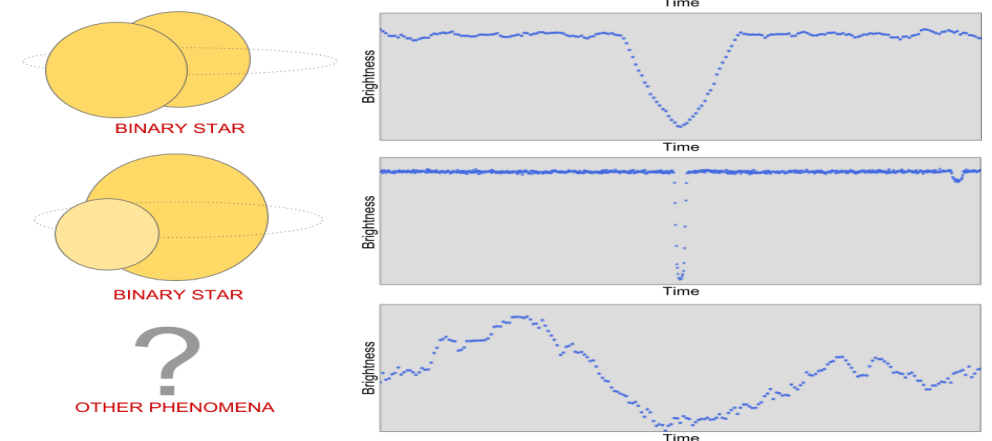
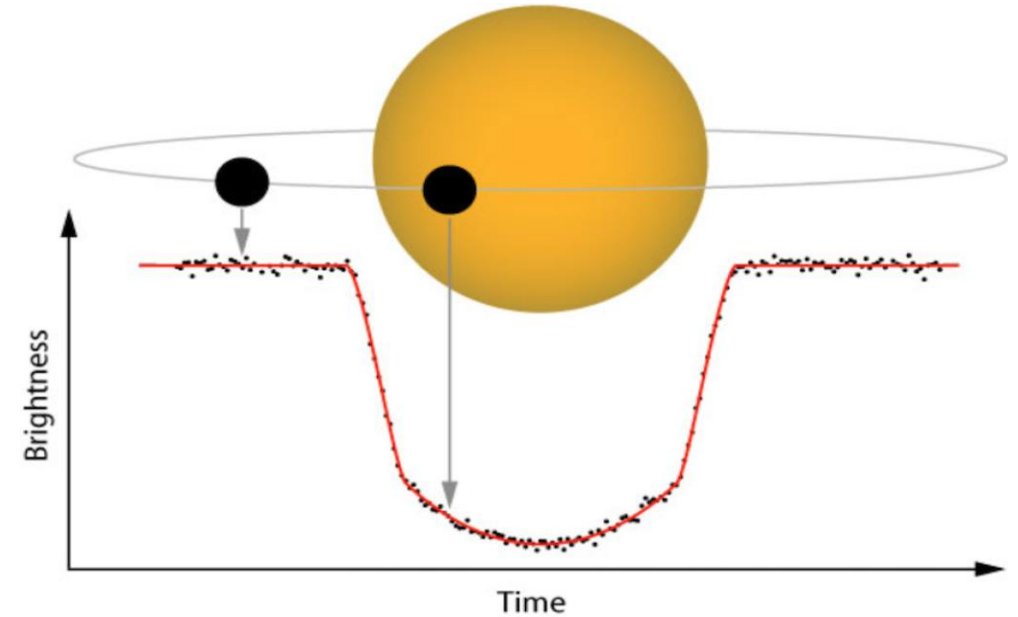
- **Photometry:** Tmag, Bmag, Vmag, JHK, WISE, GAIA
- **Astrometry:** RA, Dec, PM, Parallax
- **Stellar Params:** Teff, logg, Mass, Radius, Lum

Training Labels

Confirmed planets, false positives, eclipsing binaries from curated dataset

[Catalog Architect Link](#)

Distinguishing planet transits from stellar blends and eclipsing binaries



System Architecture

DATA LAYER | TESS MAST Archive | TIC v8.1 Catalog Cross-Match | Local Data Storage

PROCESSING LAYER | Detrending Module | BLS Period Detection | Transit Folding | Feature Extraction

AI LAYER | Training Pipeline | XGBoost / Random Forest Classifier | Cross-Validation | Model Selection

ESTIMATION LAYER | Transit Model Fitting (batman/LMFIT) | MCMC Sampling (emcee) | Parameter Extraction

OUTPUT LAYER | SNR & Confidence Scoring | Interactive Dashboard | Classification Report | 3-Page Technical Report

Tech Stack: Python 3.10+ | Lightkurve | Astropy | PyTorch/TensorFlow | XGBoost | Plotly Dash

[Detailed Layer Architect Link](#)

Technologies & Tools

Data Processing

- **Lightkurve** - TESS data access
- **Astropy** - FITS handling
- **Pandas** - Data manipulation
- **NumPy** - Numerical arrays

ML / AI Frameworks

- **XGBoost** - Gradient boosting
- **PyTorch** - Deep learning
- **scikit-learn** - ML utilities
- **Optuna** - Hyperparam tuning

Transit & Stats

- **batman** - Transit modeling
- **LMFIT** - Curve fitting
- **emcee** - MCMC sampling
- **SciPy** - Signal processing

Period Detection

- **AstroPy BLS** - Transit search
- **Lomb-Scargle** - Periodogram
- **Celerite** - GP modeling

Visualization

- **Matplotlib** - Static plots
- **Plotly Dash** - Dashboard
- **Jupyter** - Notebooks

Infrastructure

- **Python 3.10+**
- **GPU (CUDA)** - Optional DL
- **Local/Cloud**

[Tool Dependency Map Link](#)

[Github Working Prototype ML Model Link](#)

Expected Outcomes & Deliverables

- **AI Pipeline** - Automated identification and classification of periodic periodic dips in noisy TESS light curves
- **Parameter Estimates** - Orbital period, transit duration, transit depth with statistical uncertainties
- **4-Way Classification** - Confirmed Exoplanet, Eclipsing Binary, Stellar Binary, Stellar Blend, Starspot/Other
- **Confidence Scoring** - SNR-based significance levels and model confidence for each detection
- **Visualization Dashboard** - Light curves with detected signals, fitted signals, fitted models, and classification labels
- **Technical Report** - 3-page document: methodology, assumptions, assumptions, tools, uncertainty estimation



AI-powered analysis of astronomical data

[Success Metrics Dashboard Link](#)

[Evaluation Criteria Link](#)

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THANK YOU